

Some Popular MCUs Available in the Market

MCU	Key features	Interesting developments	Announced availability
LAPIS ML620530/540/550	256kB (max.) ROM, 90ch general-purpose ports, 20ch successive approximation AD converter, 8ch multifunction timer for stepper motor pulse generators, 1ch 3-phase PWM and serial communication ports for wireless ICs, peripheral MCUs	Enables wireless connection with HEMS/BEMS equipment and multichannel analogue input; newly-developed successive approximation AD converter features 1.2µs conversion speed and high accuracy ($\pm 1\%$) oscillation circuit	Samples available since February 2015 Product available since August 2015
Microchip PIC16F18877	Includes power-conserving functionality such as Idle/Doze operating modes and Peripheral Module Disable and XLP low-power technologies	Features a high level of core-independent peripherals that can be combined to perform a wide variety of functions	Available directly from Microchip
Renesas RX231	CPU core achieves a performance score of 4.16 CoreMark/MHz, 35 per cent lower than the RX200 series, only taking 0.8µA in standby mode, while retaining the contents of both SRAM and CPU registers and consuming only 120µA/MHz when the CPU is active with peripherals off	Makes it possible to substantially reduce power loss for the system as a whole during processing	Mass production beginning September 2015
Freescale S32K MCU	Includes an automotive-grade SDK, which provides middleware for the S32K's drivers. It incorporates a development platform called S32 Design Studio that includes software and design tools	Designed for software engineers, combining ARM Cortex-M4 performance with DSP functionality and a high reusability factor	Sample devices and SDKs will be available in the third quarter of 2015
Renesas RL 78/G1F	Dedicated motor control sub-system with the new auxiliary timer supports operation at up to 64MHz; software configurable, high speed rate PGA reduces cost and simplifies development; 2ch high-speed comparator enables reliable over-current detection and rotor position detection; DA converter and I2C communication function	Achieves high motor speeds with increased efficiency and reduced system costs	Varies depending on memory capacity, package and number of pins as the MCU line-up comprises 20 product versions
Atmel SMART SAM G54/G55	More SRAM, integrated sensor-fusion algorithms, multiple clock domains to enable different parts of the MCU to be individually clocked to save power	Ideal for sensor hub and battery-operated consumer applications	Available online
Marvell 88M2300 wireless MCU SoC	Fully-integrated IEEE 802.15.4 transceiver; SoC integrated with human-machine interfaces to support various MCU applications; highest RF performance with 120dBm link budget and industry-best co-existence performance; ultra-low power consumption with 8.5mA RX current and 1µA standby current; innovative XIP (execution in place) technology that supports direct code execution from flash memory	Designed to deliver improved data link quality with reduced message loss to enhance user experience	Sampling in 5x5-mm QFN-32 packages
Atmel SMART SAM DA1	Embeds PTC that offers high performance and low-power capacitive-touch support in compact QFN 5x5mm package with wettable flanks for automated optical inspection	Enhanced capacitive touch based automotive infotainment applications	Announced in February, with general sampling beginning April 2015
Broadcom BCM-58100MCU	First family of secure MCUs with embedded NFC	Integrates NFC for tap-to-access, tap-to-pair and tap-to-pay applications with a high level of security	Sampling since April 2015
Infineon XMC4800	The first EtherCATnode integrated on an ARM Cortex-M controller with on-chip flash and analogue/mixed-signal capabilities	Enables the most compact designs, eliminating the need for a dedicated EtherCAT ASIC, external memory and crystal	Samples available since August 2015; volume production starts from Q1 2016
Renesas RX23T Group	Built-in floating-point processing unit; current consumed in software standby mode (with RAM retention) is 0.45µA	Precise control and quieter motor in home appliance	Mass production since August 2015
STMicroelectronics SPC58NExx	MCUs compliant with automotive safety standards, encryption for security and increased memory size for the storage of vital programs and data	Fault-tolerant MCUs for demanding applications throughout the car	Sampling since April 2015
STMicroelectronics STM32F469/479	ARM Cortex-M4 MCU (MCU unit) core augmented by ChromART Accelerator and ART Accelerator that speed up graphics and real-time processing as well as access to the MCU's program memories	MIPI display serial interface has been integrated in addition to the TFT-LCD controller; LCD-TFT and MIPI-DSI display interfaces enable an advanced UI while granting enough resources for demanding real-time processing	Sampling since April 2015
STMicroelectronics STM32L4	At an active power consumption as low as 100µA/MHz, this IC can achieve a performance of up to 100DMIPS	Optimised for ultra-low-power consumption for wearable devices	General release was in Q2 2015
TI RF430FR15xH	NFC sensor transponder for industrial, medical, wearables and IoT applications	Industry's first highly-integrated FRAM MCU	Already available
TI MSP432P401R	Industry's lowest power ARM Cortex-M4F secure MCU	ULPBench score of 167.4 (850nA standby, 95µA/MHz active)	Already available
TI MSP430FR4133	Integrated LCD with on-chip charge pump and up to 60 GPIO/capacitive touch I/O	Ultra-low-power FRAM MCU	Already available

There is also a new generation called CAN Flexible Data rate (CAN FD) with variable data rate," explains Girish K., project manager, Embitel.

Trying to serve the need of high motor speeds with reduced system

costs, Renesas has recently launched a group of multifunction MCUs that support Infrared Data Association (IrDA) serial communication functionality for devices, such as simplifying dishwashers and refrigerators,

diagnostics, monitoring, e-locks or handy terminals. Talking about terminals, Broadcom's BCM58100 MCU integrates NFC for tap-to-access, tap-to-pay and tap-to-pay applications.

Infineon XMC4800 claims to be